

An Introduction to Graphviz

Graphviz comprises a very flexible and handy set of tools that is freely available under an open source licence. Read on to get more familiar with it.



Graphviz tools help you draw, illustrate and present graph structures. Do not be discouraged and please do not think that 'drawing graph structures' is restrictive and limiting. I can promise you that by the end of the article, you will change your mind.

The good thing is that Graphviz algorithmically arranges the graph nodes so that the output is both practical and pleasing! Graphviz can be used in domains such as software engineering, networking, bioinformatics, databases, Web structures and knowledge representation. The central part of Graphviz consists of implementing algorithms for graph layouts. Most Graphviz code is written in C.

Installing Graphviz

Your Linux distribution probably includes a ready-to-install Graphviz package, so go ahead and install it, after which you can try to compile the following code:

```
digraph G
{
    "Hello world!";
}
```

Use the following command for the compilation:

```
$ dot -Tps hw.dot -o hw.ps
```

If you see no error messages, then you are ready to continue reading the rest of the article. The aforementioned command will produce a Postscript file called *hw.ps* that you can view. There are more command line arguments that will not be discussed here. You can check the man pages or the website for the tools for more information.

Note that the word *digraph* means that a directed graph is going to be created. To create an undirected graph, the word *graph* is used instead. It is easily understood that for such a simple example, it does not make any difference if the graph is either directed or undirected.

What is Graphviz?

Graphviz (or GraphViz or graphviz) is a collection of tools for manipulating graph structures and generating graph layouts. Graphviz supports either directed or undirected graphs. It offers both graphical and command line tools but in this article, the command line tools will be used. A Perl-to-Graphviz interface library is also available, but it is not covered in this article. There is also a C++ interface.

Strictly speaking and according to the book 'The Design and Analysis of Computer Algorithms', a graph $G=(V, E)$ consists of a finite and non-empty set of vertices V and a set of edges E . If the edges are ordered pairs of vertices, then the graph is said to be *directed*. If the edges are unordered pairs, the graph is said to be *undirected*.